

Vicente Arroyos

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EDUCATION

University of Washington, Seattle - 2016–2026

- Paul G. Allen School of Computer Science & Engineering
 - Ph.D., (graduated 12/18/2026)
 - Master of Arts and Science (graduated 06/2022), GPA: 3.9/4.0
- Electrical & Computer Engineering Department
 - Bachelor of Science: Embedded Systems, Minor in Mathematics (graduated 05/2020), GPA: 3.4/4.0

Queensland University of Technology - Summer 2018

- Aeronautics & Astronautics, Australia. Design of novel materials and structures program. GPA: 4.0/4.0
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AWARDS & FELLOWSHIPS

Graduate

- Clean Energy Institute Graduate Fellowship (2023)
- Washington NASA Space Grant Consortium Fellowship (2021–2024)
- SPEEA ACE Fellowship (2023)
- National GEM Consortium Fellowship (2021)

Undergraduate

Ronald E. McNair Postbaccalaureate Achievement Program Scholarship (2019), Vincent Kerkof Endowed Scholarship (2019), James M. & Virginia C. B. Webster Endowed Scholarship (2019), NACME Scholar (2018, UW Dean's List (2018), WSOS Scholar (2016), UW Brotherhood Initiative (2016–2020), OMAD UW LSAMP (2016–2020)

ACADEMIC & INDUSTRY POSITIONS

Iyer Lab, University of Washington - 2021–Present Advisor: Vikram Iyer | Graduate Research

- Microrobotics & Edge AI: Co-designed MilliMobile and Coin-Copter platforms; integrating TinyML visual inference to transform sub-gram/millimeter-scale robots into autonomous edge agents within milliwatt power envelopes.
- Sustainable Hardware: Established design rules for biodegradable electronics to create circular robotic systems that reduce electronic waste.
- Fabrication & Control: Engineered onboard power, sensing, and control systems for insect-scale origami microfliers with shape-changing airframes for controlled wind-borne dispersal.
- Local-First AI: Developed CourseSLM, a privacy-first AI scaffolding tool running locally on edge devices (Apple MLX) to support K-12 computing education in internet-constrained environments.

NSF Civic Innovation Challenge, University of Washington - 2024–2026 "Leveraging a Connected Network of Unattended Micro Pantries" (Iyer/Goodchild/Spiker/Hovis teams + civic partners)

- Designed ESP32-based sensing platform fusing thermal probes, load cells, and ambient sensors for real-time supply, demand, and food safety monitoring.
- Implemented secure intermittent offload protocol to Azure IoT with local-first logging during outages.
- Scaling deployment to 8 sites with real-time telemetry dashboard.

NSF Smart Agriculture, University of Washington - 2024–2025 "Community Based AI for Black Diasporic Farming" (Iyer/Rosner/Ghoshal groups)

- Architected a local-first sensor network for data sovereignty, enabling communities to steward reclaimed land without reliance on cloud systems.
- Built sensing stack (Raspberry Pi, Modbus & GPIO sensors for CO₂/Soil/Wind) with LoRa network for long-range telemetry and Flask/SQLite local dashboard.
- Deployed and tested data persistence across power cycles and network outages.

Networks & Mobile Systems Lab, University of Washington - 2020–2021 Advisor: Shyam Gollakota | Graduate Research

- Designed low-power battery-free robotic systems using bistable leaf-out origami. Implemented A3C & PPO reinforcement learning to define efficient folding patterns.

Laboratory for Engineered Materials & Structures, University of Washington - 2018–2020 Advisor: Jinkyu Yang | Undergrad Research

- Demonstrated that leaf-out origami can kinematically output 150% of energy input. Fabricated prototypes and proved analytically that tailored designs vary normalized energy outputs.

Oak Ridge National Lab, Neutron Sciences - Jun–Aug 2021 Mentor: Rob Knudson | GEM Fellow Intern

- Processed 8000+ curves (100K+ points each) to extract 11 features for gamma/neutron classification.

Pacific Northwest National Lab (PNNL) - Jun–Aug 2018 DOE SULI Intern

- Developed CyPhyTown (Raspberry Pi cyber-physical system) for outreach on cyber-physical security.
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PUBLICATIONS

Peer Reviewed Journal Articles

1. Arroyos, V., Johnson, K., Donahoe, G., Njie, O., Yewondwossen, S., Arroyos, R., Fuller, S., and Iyer, V. "A Stable Coin-sized Helicopter Design with Enough Lift for Autonomy" *Science Advances*, in Revision, March 2026.
2. Arroyos V., Johnson K., Elberier T., Fuller S., Iyer V., and Gollakota S. "Solar-powered Shape-changing Origami Microfliers." *Science Robotics*. 8, eadg4276 (2023).
3. Yasuda H., Johnson K., et al. "Leaf-like Origami with Bistability for Self-Adaptive Grasping Motions." *Soft Robotics*. (2022).

Peer Reviewed Conference Papers

1. Arroyos V., Cora A., Kimanje A., et al. "Tool or Toy? Student Views on ChatGPT in Culturally Responsive Computing Education." *WEEF-GEDC*, Daegu, Republic of Korea (2025).
2. Arroyos V., Johnson K., Hussein L., et al. "Scalable Community Mentorship: A Vision for Engineering Literacy & Access." *WEEF-GEDC*, Sydney, Australia (2024).
3. Cora A., Arroyos V., Hussein L., et al. "Leveraging AI to Improve STEM Engagement for Black and Latine Youth." *BICE*, Santo Domingo, Dominican Republic (2024).
4. Arroyos V., Johnson K., Englhardt Z., Yin D., Patel S., and Iyer V. "MilliMobile: An Autonomous Battery-free Wireless Microrobot." *ACM MobiCom* (2023).
5. Arroyos V., Johnson K., Villanueva R., Schulz A., Fuller S., and Iyer V. "Toward Sub-Gram Helicopters: Designing a Miniaturized Flybar for Passive Stability." *IEEE/RSJ IROS* (2023).
6. Arroyos V., Viitaniemi M., Keehn N., et al. "A Tale of Two Mice: Sustainable Electronics Design and Prototyping." *ACM CHI EA* (2022).

In Preparation / Submitted

1. Johnson K., Arroyos V., Cardenas K., Barrow I., Lyu J., and Iyer V. "FLEABOT: a Field Locomotive Electromagnetic Actuated roBOT." (in preparation).

Patents

1. Johnson K., Arroyos V., Donahoe G., Fuller S., and Iyer V. "Coin-copter: Battery-free Near-gram Helicopter Designs." U.S. Provisional Patent Application (2025).
2. Johnson K., Arroyos V., Englhardt Z., Yin D., Patel S., and Iyer V. "Wireless microrobot and operating techniques." International Patent Application WO 2025/058992 A1 (2024).
3. Johnson K., Arroyos V., Villanueva R., Schulz A., Fuller S., and Iyer V. "Flybar apparatus." International Patent Application WO 2024/137785 A1 (2023).

Refereed Posters & Workshops

1. Arroyos V., Ibrahim M., Mensah E., Johnson K., Fuller S., Iyer V. "Battery-free Computer Vision on Insect-scale Microrobots." *IEEE ICRA Late Breaking* (2025).
 2. Johnson K., Arroyos V., et al. "AVELA: Understanding Why Technology Alone Is Not Enough." *arXiv Preprint* (2024).
 3. Johnson K., Arroyos V., Yang J. "Leaf-out Origami: Tailorable Fabrications and Applications." *UCLA Ronald E. McNair Conference* (2019).
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TEACHING

University of Washington

- CSE474: Introduction to Embedded Systems (Spring 2023) - Lead TA
- Engineering Peer Educator, GEN ST 199 (2019)
- UW Math Academy Tutor - Python, Arduino labs (2019–2020)

AVELA Curricula

- 60-hour Introduction to Embedded Systems - adopted in 15 classrooms (2020–2023)
 - 40-hour Introduction to Python & Machine Learning - adopted in 30 classrooms (2021–2022)
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MENTORSHIP

Graduate: Colin Kessler (Edinburgh Ph.D.), Laila Walker (UW CSE Ph.D.), Michael Ibrahim (UW CSE M.S.), Tilboon Elberier (UCLA ECE Ph.D.)

Undergraduate (17 students): Including Giannah Donahoe (now Microsoft), Kevin Hernandez-Ramos (now Duolingo), William Vera (now Meta), Raul Villanueva (now Kenworth Trucks)

GRANTS & GIFTS

External (AVELA Co-Founder)

- Apple Donations (2024–2026): \$155,000
- BECU People Helping People Award (2024): \$50,000
- SPEEA ACE Contracts (2023–2026): \$255,000
- Amazon Future Engineers (2022–2026): \$651,000

Government

- Department of Energy, Water Power Technologies Office (2025–2027): \$33,000
- WA King County 4Culture (2024–2026): \$144,000

University of Washington

- UW Department Grants (2019–2025): \$150,000
 - UW Student Technology Fee (2024): \$65,000
 - Washington NASA Space Grant (2023): \$13,000
 - ISCRM (2021–2025): \$40,000
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PRESENTATIONS

Japan Aerospace Exploration Agency (JAXA) (2024), Keynote, Upward Bound Research for Public Good (2022), Panelist, Ronald E. McNair Fellowship Workshop (2022), UW Molecular Information Systems Lab (2022), Panelist, Gabriel E. Gallardo Research Student Leadership & Advocacy Symposium (2022), UW STARS Undergraduate Seminar (2020), Panelist, Brother Initiative Workshop (2020), UW Ronald E. McNair Conference (2019), Panelist, McNair Upward Bound Workshop (2019)

ENTREPRENEURSHIP

AVELA - A Vision for Engineering Literacy & Access (Co-Founded 2022)

- Nonprofit (EIN: 88-4125697) promoting economic opportunities in STEM for underserved students.
- 40+ research intern positions, one annual scholarship, 6 refereed publications.
- 500+ members partnered with ~100 faculty, schools, and community partners to teach 6,000+ students (110,000+ STEM instruction hours).

NSF I-Corps

- UW CoMotion: "Designing a Flybar for Stabilizing Millimeter-scale Flying Robots" (2023)
 - MIT Spark: "AVELA - A Vision for Electronic Literacy & Access" (2022)
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SELECTED MEDIA COVERAGE

- Ridwell Blog, "Micro-pantries, big science and Ridwell is part of it" (2026)
 - UW CSE, "Self Powered Mini Robot" (Dec 2023)
 - UW CSE, "Battery-free origami microfliers offer a new bio-inspired future of flying machines" (Sept 2023)
 - CNET, "These Tiny Sensors Ride the Wind Like Dandelion Seeds" (May 2022)
 - UW News, "Q&A: Making Earth friendly electronics" (Apr 2022)
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SERVICE

- AVELA Board Treasurer (2022–Present)
- UW CSE Recruiter: SHPE, Tapia, CRA WP conferences (2022–2025)
- UW LSAMP Ambassador (2018–2020)